KUNDAN KUMAR MISHRA

PhD Candidate, Bioengineering | The University of Texas at Dallas
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EDUCATION

PhD in Bioengineering Jan 2023 – Present
University of Texas at Dallas, TX, USA

- GPA: 4.0 / 4.0

M.Tech in Biomedical Engineering 2017 – 2019
Indian Institute of Technology Ropar, India

- CGPA: 7.58 / 10

B.E. in Electronics and Instrumentation 2010 – 2014
Hindustan Institute of Technology and Science, Chennai, India

- CGPA: 8.44 / 10

RESEARCH INTERESTS

Electrochemical sensor development · Assay development · Microfluidics · Microbiology · Bacterial culture · Biosensors · Sensor fabrication

PROFESSIONAL EXPERIENCE

PhD Research Fellow – Bioengineering Jan 2023 – Present
University of Texas at Dallas, TX, USA

- Developed an intelligent multiplexed electrochemical impedance spectroscopy (EIS) biosensor for simultaneous detection of antibiotics and pesticides in complex food matrices.
- Validated sensor performance across diverse food and environmental matrices (chicken, shrimp, milk, salad, agricultural soil run-off) through spike-and-recovery and reproducibility studies.
- Designed and fabricated sensors using screen-printing techniques; executed full electrochemical testing and validation pipeline.
- Developed nano-scale assays for detecting pathogens, pesticides, toxins, and antibiotics.

Junior Research Fellow Jan 2022 – Dec 2022
Indian Institute of Technology Delhi, India

- Contributed to the Nano-electronics Network for Research and Application (NNetRA) program.
- Investigated non-silicon-based nano-fabrication technologies and nano-scale device applications.
- Designed and developed microfluidic lab-on-a-chip nanobiosensors for detection of emerging viral pathogens.

Research Associate Jun 2019 – Oct 2021
Virtual Sense Global Technologies, Pune, India

- Conducted extensive research on molecular diabetic biomarkers in human breath.
- Functionalized multiwalled carbon nanotubes (MWCNTs) with COOH, OH, and NH3 groups via distillation and vacuum filtration.
- Applied a peroxidase enzyme-based approach to improve biosensor specificity.

- Designed circuit and sensing setups for gas sensing experiments; calibrated commercial sensors (humidity, CO2, gas) to mimic in-vivo conditions.
- Modelled and prototyped the sensing system using 3D printing.
- Conducted clinical trials: hypothesis development, methodology, pilot study, data collection, and analysis.

Teaching Assistant – Introduction to Medical Products (Electrical Dept.) Aug 2018 – Nov 2018

Indian Institute of Technology Ropar, India

- Assisted Prof. Brajesh Rawat in delivering undergraduate coursework in medical product engineering.

TECHNICAL SKILLS

Bio-Lab: DNA/RNA/Plasmid/Protein isolation, bacterial culture handling, BSL-II, autoclave sterilization, BCA assays, PCR, RT-PCR, Western Blot, ELISA, downstream processing, protein purification, IPTG induction

Biosensors: Electrochemical sensor development, biosensing, cleanroom microfabrication, photolithography design and optimization

Design/CAD: FreeCAD, Autodesk Fusion 360, Autodesk Eagle, COMSOL

Programming: Python, MATLAB, Arduino

PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Mishra, K.K**., Dhamu, V.N., Kokala, A., Muthukumar, S.; Prasad, S., Advancing Food Safety: Two-Plex Electrochemical Biosensor for Mycotoxin Detection in Food Matrices, Biosens Bioelectron X, vol. 25, Sep. 2025
2. **Mishra, K.K.,** KM Thakkar, Dhamu, V.N, Muthukumar. S, Prasad.S , Portable Electrochemical Sensing Platform for Aflatoxin B1 Detection in Food Matrices Food Analytical Methods 19 (1), 19
3. **Mishra, K.K**., Dhamu, V.N., et al. Electrochemical Sensor Platform for Rapid Detection of Foodborne Toxins," Biosensors (Basel), vol. 15, no. 6, p. 361, Jun.2025.
4. **Mishra, K.K**.; Dhamu, V.N.; Jophy, C.; Muthukumar, S.; Prasad, S. Electroanalytical Platform for Rapid E. coli O157:H7 Detection in Water Samples. Biosensors 2024, 14, 298.
5. **Mishra, K.K**., Dhamu, V. N,; Muthukumar, S.; Prasad, S Quick and Sensitive Two-Plex Electrochemical Platform for Pathogen Detection in Water., Nano Select 6 (11), e70017
6. **Mishra, K.K**.,, Thakkar, KM,' Karmakar, S.; Dhamu, V.N,; S Muthukumar, Prasad, S. Duplex EIS Sensor for Salmonella Typhi and Aflatoxin B1 Detection in Soil Runoff. Biosensors 15 (10), 654
7. Spiro, J C*.; **Mishra, K.K***.; Dhamu, V.N.; Bhatia, A.; Muthukumar, S.; Prasad, S. Development of a portable electrochemical sensing platform for impedance spectroscopy-based biosensing using an ARM-based microcontroller. Sensors & Diagnostics 2024
8. S Karmakar, D Poudyal, **KK Mishra**, VN Dhamu, S Muthukumar, S Prasad, Label-Free Electrochemical Biosensor for Real-Time Detection of Live Salmonella typhimurium in Salad Samples Using Non-Faradaic EIS, Biosensors and Bioelectronics, 2025
9. P Singh, A Kumar, **KK Mishra**, KC Lin, S Muthukumar, S Prasad, Temporal Profiling of Cytokines in Passively Expressed Sweat for Detection of Infection Using the AWARE Device, ACS sensors, 2025
10. P Singh, A Kumar, MZ Ikram, **KK Mishra**, KC Lin, S Muthukumar, S Prasad Development and benchtop testing of an aptamer-based sweat sensor for CRP detection, Microchimica Acta 192 (11), 1-13
11. B Rawat, **Mishra, K.K**,. U Barman, L Arora, D Pal and Roy P. Paily, Two- Dimensional MoS_2 -based Electrochemical Biosensor for Highly Selective Detection of Glutathione, Accepted to IEEE Sensor Journal, 2020 **Mishra, K.K**., Dhamu, V.N., Poudyal, D.C. et al. PathoSense: a rapid electroanalytical device platform for screening Salmonella in water samples. Microchim Acta 191, 146 (2024).
12. Introduction to Bio-Nanotechnology- book chapter

13. DC Poudyal, VN Dhamu, M Samson, K Mishra, D Baby, S Muthukumar, Lateral Flow Based Immunosensors for Monitoring Food Quality and Safety, The Royal Society of Chemistry, 2025- book chapter.

Book Chapters

1. Mishra, K.K. Introduction to Bio-Nanotechnology (book chapter).
2. Poudyal, D.C., Dhamu, V.N., Samson, M., Mishra, K., Baby, D., Muthukumar, S. Lateral Flow Based Immunosensors for Monitoring Food Quality and Safety. The Royal Society of Chemistry, 2025.

SELECT RESEARCH PROJECTS

Functionalization and Binding Analysis of MoS2 Nanoparticles with GST Proteins

- Characterized GST-MoS2 composite using FTIR, XRD, FESEM, Raman, and AFM.
- Calculated sensitivity and limit of detection (LOD) across a wide linear detection range.

Breast Cancer Diagnosis via Potassium Ion Sensing

- Functionalized MoS2 nanoparticles with single-stranded DNA and performed binding analysis.
- Prepared chemiresistive sensor for electrochemical disease diagnosis.

Mycobacterium tuberculosis Diagnosis from Breath

- Functionalized graphene oxide nanomaterials with metal ions.
- Measured preliminary sensor response in vitro.

HONORS & AWARDS

- ECS Jonsson Family Bioengineering Graduate Fellowship (2024–2025)
- Convergence Grant 2025
- ECS Fall 2025 Travel Award
- 2025 BMES Annual Meeting Travel Grant

PRESENTATIONS & WORKSHOPS

- Poster: "A Rapid Electroanalytical Device Platform for Screening Salmonella in Water Samples," BMES 2024, Baltimore, MD.
- Poster: "Highly Sensitive Detection of Glutathione Based on Layered MoS2 Electrochemical Sensor," Research Conclave 2019, IIT Ropar.
- Invited Talk: "How Electrochemical Sensors Aid in Glutathione Detection," IIT Ropar.
- Attendee: UK-India Interdisciplinary Workshop on Advanced Biomaterials and Biosensors, IIIT Ropar.
- Attendee: Workshop on Generation of Electricity by Wind Energy.